

A dated, falsifier-equipped comparison of a fixed-point constant with the neutrino mass-squared ratio

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preprint, not peer reviewed, version 2026-06-11 (supersedes the 2026-06-10 internal registration; the changes from that version — including a self-consistency correction that *reduces* the claimed tension — are listed in Appendix A).

Abstract

We document, with falsifiers and an honest trials accounting, a comparison between a dimensionless constant arising in a stochastic-process program for fundamental physics and the neutrino mass-squared splitting ratio. The constant $\varepsilon = 3\kappa - 1 = 0.031768636446582$ derives, with no Standard-Model input, from the unique real root of $\theta^3 + \theta^2 + \theta = 1$ (the reciprocal tribonacci constant — an *algebraic* number whose identities $\tanh \eta = e^{-\eta}$, $\theta = e^{-\eta}$, and $\tanh(\eta/2) = \theta^2$ are elementary and verified) through the binding analysis now fully self-contained in the submitted companion [C0]; κ is defined in Section 1. **What is registered — rebuilt after two independent hostile reviews found the previous version’s registered claim incoherent, and we agree.** The earlier version registered “the exponent $x_\nu = 1/2$ ” while simultaneously asserting the program does not determine the $O(1)$ coefficient C in $m_2/m_3 = C\varepsilon^{1/2}$. Those cannot coexist: with one observable, coefficient and exponent are exactly degenerate ($x = 1/2 + \ln C / \ln \varepsilon$), and the program’s own 68% texture spread $C \in [0.22, 1.26]$ covers an exponent interval of width 0.506 — more than the rung spacing 0.5, i.e. *a lattice with free $O(1)$ coefficients covers the entire exponent axis and has zero discriminating power*. The registered claim is therefore the strictly stronger statement the program motivates but does not derive: the **undressed spectrum point** $m_1 : m_2 : m_3 = \varepsilon : \sqrt{\varepsilon} : 1$ exactly (no texture dressing in the neutrino sector), giving the self-consistent prediction $\sqrt{\Delta m_{21}^2 / \Delta m_{31}^2} = \sqrt{\varepsilon / (1 + \varepsilon)} = 0.175472$ (exact ε ; see the Section 2 precision correction) against the registration-snapshot fit 0.1718 ± 0.0026 : a 1.4σ tension at registration. **The expected outcome is stated against our own interest, at both reference centrals:** at the snapshot central, JUNO-class precision ($\sigma_S \approx 0.00067$) excludes the registered point at $\approx 5.5\sigma$ — the kill zone; at JUNO’s own first measurement (Appendix B: $S = 0.17283$), the expectation is $\approx 4.0\sigma$ — inside the pre-assigned $(2\sigma, 5\sigma]$ *failed* band rather than the kill zone. Either way the maximum-likelihood extrapolation is that this claim *fails*; we register it anyway because the program asserts it, both outcomes are informative, and a registration that hides its own expected failure is theater. The trials accounting is given in full and corrected (the earlier version misstated its own record: under the band-grade criterion the charged ratio m_e/m_μ , at exponent 1.546, *is* a hit of rung 1.5). The spectrum consequence $\Sigma m_\nu \approx 0.0607$ eV differs from the normal-ordering floor (0.0587 eV) by only ~ 1.9 meV, below projected cosmological sensitivity — a consistency statement, not a discriminating test. Voiding

scenarios, including inverted ordering, are spelled out, and the running escape hatch is closed with a computed bound. Full disclosure up front: the program separately holds a mechanism-derived *successor* claim set (a dressed four-member menu with two members live, one of them at -0.05σ against the JUNO first data) — cross-declared in the body, with the burden transfer stated, so this note’s expected failure cannot be read as costless.

1. Provenance and definition of the constant

The constant arises in a research program deriving quantum and gauge-theoretic structure from stochastic processes with sealed records (the program papers, including the mathematical companions with full proofs, accompany this note as a citable batch). The chain, with its load-bearing identifications named:

1. $\theta = 0.5436890\dots$ is the unique real root of $\theta^3 + \theta^2 + \theta = 1$: an **algebraic** number (the reciprocal of the tribonacci constant), produced in the program as the fixed point of projective consistency under binary refinement of records. The companion identities ($\tanh \eta = e^{-\eta}$ with $\theta = e^{-\eta}$; $\tanh(\eta/2) = \theta^2$) are elementary algebra, stated because all three forms occur in the program’s renormalization analysis.
2. κ is the **single-relation first-order marginality coefficient** of the binding theory — the coefficient in the exact law $\theta^w(1 - w\kappa)$ — computed (not fitted) in closed form, $\kappa = \eta(1 - \theta^2)/(1 - \theta^2 + \theta) = 0.343922878815527$ (the submitted companion [C0], with exact algebraic relations), giving $\varepsilon = 3\kappa - 1 = 0.031768636446582$; all comparisons below use the exact value (Section 2’s precision correction records the earlier 3-digit rounding).
3. The provenance predates any neutrino comparison by a long, internally dated sequence of program papers. *Honesty note*: the numerical proximity to the neutrino ratio was noticed before the internal registration and motivated it. What this note adds: precise claim forms, tolerances, falsifiers, the trials accounting, and the correction of Appendix A. We make no priority-of-prediction claim for the central-value agreement — only the forward tests below count.

2. The observable and the self-consistent prediction

Under normal ordering with the program’s spectrum $m_1 : m_2 : m_3 = \varepsilon : \sqrt{\varepsilon} : 1$:

$$\sqrt{\frac{\Delta m_{21}^2}{\Delta m_{31}^2}} = \sqrt{\frac{m_2^2 - m_1^2}{m_3^2 - m_1^2}} = \sqrt{\frac{\varepsilon - \varepsilon^2}{1 - \varepsilon^2}} = \sqrt{\frac{\varepsilon}{1 + \varepsilon}} = 0.175472.$$

Precision correction (found in review). The registration snapshot computed this value at 3-digit $\varepsilon = 0.0318$, printing 0.17556; at the exact constant ($\varepsilon = 0.031768636446582$, companion [C0]) the registered point is **0.175472** — an 8.4×10^{-5} shift that is 0.13σ at this note’s JUNO-class precision ($\sigma_S = 0.00067$; it is 0.23σ at the ~ 0.00037 ultimate-precision convention used elsewhere in the program — both stated to avoid a units trap). The registered *claim* (the undressed spectrum point) is unchanged; its exact evaluation is 0.175472, used everywhere below; the rounded print survives only in this paragraph as the correction record.

The naive hierarchical formula $\sqrt{\varepsilon} = 0.17824$ ignores m_1 and errs by $O(\varepsilon) \approx 1.6\%$ — comparable to the experimental precision, so the self-consistent form is mandatory. Current global-fit inputs (NuFIT 5.x-class, normal ordering, without SK-atmospheric: $\Delta m_{21}^2 = (7.41 \pm 0.21) \times 10^{-5} \text{ eV}^2$, $\Delta m_{31}^2 = (2.511 \pm 0.027) \times 10^{-3} \text{ eV}^2$; the mild fit correlation between the two splittings is neglected, which slightly *overstates* our error bar):

$$S_\nu = 0.1718 \pm 0.0026, \quad \frac{S_\nu}{0.175472} = 0.979, \quad \text{deviation} = 1.4\sigma.$$

3. The claims, with the trials accounting first

Trials accounting (read before the claims; corrected twice over in this version). The program’s half-integer exponent lattice spaces rungs 0.5 apart in $x = \ln S / \ln \varepsilon$. All distances are measured to the *self-consistent* prediction (exponent $x_{\text{pred}} = 0.5045$, not the bare rung 0.5 — an earlier version mixed the two targets; using one target consistently *strengthens* the agreement: $|x_\nu - x_{\text{pred}}| = 0.0062$). (i) The probability that a random exponent lands within the observed distance of the prediction is $2(0.0062)/0.5 = 2.5\%$ per trial (fragile: the distance 0.0062 carries its own measurement uncertainty $\sigma_x \approx 0.0044$, so this per-trial figure is indicative, not sharp). (ii) A band-tolerant criterion ($|x - \text{rung}| \leq 0.065$, the withdrawn band’s width) costs 25.9% per trial. (iii) The available RG-clean lepton-sector ratios number at least three (m_e/m_μ , m_μ/m_τ — pole masses, QCD-free — and S_ν); if the lattice is read as sector-wide (the charged-quark steps motivated it), the quark-sector ratios belong in the band-grade denominator too, which would further dilute the central-value evidence — counted against us either way. **Correction of our own record:** the earlier version stated the charged ratios ”do not land on rungs” (exponents 1.546 and 0.819) — but under the band-grade criterion of the same paragraph, m_e/m_μ at 1.546 is a hit of rung 1.5 ($|\Delta x| = 0.046 \leq 0.065$). So the honest record is: band-grade, 2 of 3 trials hit (chance of ≥ 2 accidental band-grade hits in 3 trials: $\approx 16.6\%$); observed-grade, 1 of 3 (≥ 1 accident: $\approx 7.3\%$). (iv) *Constant-selection freedom, counted against ourselves:* the program’s small-constant menu ($\theta = 0.544$, $\theta^2 = 0.296$, $\theta^3 = 0.161$ — itself within 7% of S_ν — κ , η , ε , $\sqrt{\varepsilon}$) offers post-hoc matching freedom; the defense is that the base ε and the half-integer lattice were fixed by the *charged* sector years before the neutrino comparison, but a skeptic discounts post-hoc readings for menu freedom, and we agree. (v’) *Registered-claim multiplicity, counted:* program-wide there are three live registered points (this note’s, plus the two live members of [IX]’s four-member menu); a skeptic should treat the ensemble as three trials on one observable, which further dilutes any central-value credit — counted against us, like everything else in this section. (v) *The dressing problem, stated as the structural lesson:* the same $O(1)$ Yukawa dressing invoked for the charged misses would, if available in the neutrino sector, absorb any outcome — which is why the only registrable claim is the undressed point below, and why the central-value agreement today carries **little evidential weight**: everything of value is forward-looking.

Claim S (the registered content: the undressed spectrum point). The program asserts the texture-free spectrum $m_1 : m_2 : m_3 = \varepsilon : \sqrt{\varepsilon} : 1$ exactly, hence $S_\nu = \sqrt{\varepsilon}/(1 + \varepsilon) = 0.175472$ (exact ε ; Section 2) up to the experimentally-shrinking error. This is a *point* claim: it is the program’s spectrum, it is falsifiable, and it is strictly stronger than the exponent-lattice statement (which clause (v) shows is unfalsifiable while C is free). **Expected outcome, computed and pre-committed:** JUNO ($\sigma_{21} \approx 0.6\%$, $\sigma_{31} \approx 0.5\%$, hence $\sigma_S \approx 0.00067$) yields a deviation of $\approx 5.5\sigma$ at the registration-snapshot central ($\approx 4.0\sigma$ at the JUNO-first central,

Appendix B) — the claim is *expected to fail* on either reading. Pre-committed readings at JUNO-final precision: within 2σ = survives (requires the central values to drift); in $(2\sigma, 5\sigma]$ = the registration **fails** its survival criterion and we report it as failed (revivable only by subsequent central-value drift, and we will not relabel failure as "tension"); beyond 5σ = the undressed spectrum point is dead, we say so in an update, and what dies with it is the program's *no-dressing assertion for the neutrino sector* — the exponent lattice itself survives only in the unfalsifiable free- C form, which we do not register.

Consequence (consistency, not test). $\Sigma m_\nu = m_3(1 + \sqrt{\varepsilon} + \varepsilon) \approx 0.0607$ eV — only ~ 1.9 meV above the massless- m_1 normal-ordering floor (0.0587 eV), below projected cosmological sensitivity. Recorded for completeness; we claim no discriminating power.

What we do not register. (a) No coefficient band: the program's own analysis proves the $O(1)$ coefficient is texture, not structure (it spreads over $[0.22, 1.26]$ at 68% under unconstrained textures), so any band narrower than that spread would be drawn around the known value, not derived — the earlier internal version's $[0.80, 1.25]$ band is withdrawn for exactly this reason. (b) No exponent-lattice claim with free coefficient: zero discriminating power, per the abstract's computation.

4. Premises and voiding/falsifying scenarios

- **(h) Hierarchy/ordering premise.** Normal ordering with the program spectrum. *Voided by:* established inverted ordering (JUNO/atmospheric — listed explicitly as a voiding outcome); established quasi-degeneracy from cosmology or endpoint measurements. Voiding is recorded as premise failure, not as a pass.
- **F-S:** JUNO-era exclusion of $S_\nu = \sqrt{\varepsilon/(1 + \varepsilon)}$ at $> 5\sigma$ kills the undressed spectrum point; the $(2\sigma, 5\sigma]$ band is pre-assigned as *failed* (Section 3) — no outcome is left unassigned.
- **Running, bounded now (closing the escape hatch an earlier version left open).** The comparison is made at the low-energy oscillation scale. Under Standard-Model running (no extended Higgs sector), the flavor-dependent part of the neutrino mass-ratio evolution is governed by the τ Yukawa: $|\Delta(m_2/m_3)|/(m_2/m_3) \lesssim (y_\tau^2/16\pi^2) \ln(\Lambda/m_Z) \cdot O(1) \approx 1.04 \times 10^{-4}/158 \times 23 \times O(1) \approx 5 \times 10^{-5}$ for matching scales up to $\Lambda = 10^{12}$ GeV — four orders below JUNO precision. The registration is pinned to this: SM running, $\Lambda \leq 10^{12}$ GeV, declared now. A future exclusion cannot be attributed to running within these premises; invoking an extended sector (e.g. large- $\tan \beta$ two-doublet running, which can reach percent level) would itself be a premise change and is pre-declared as such, not as a rescue.

5. What this note is

A dated, falsifier-equipped statement of a forward test whose *expected outcome under current data is its own death* — stated plainly, with the trials mathematics computed against our own interest and successive corrections of our own claims placed in the open. Three honesty items that bound what this note is worth: (1) the constant's derivation chain lives in the program corpus, which accompanies the submission as supplementary material but is not part of the peer-reviewed batch — "checkable" means checkable-given-supplement; (2) **a timing error in an**

earlier draft, corrected: that draft claimed this registration predates JUNO’s first oscillation-parameter release — false; JUNO’s first measurement (59.1 days of data) appeared in November 2025 (arXiv:2511.14593), *before* this note, and the draft moreover quoted superseded NuFIT 5.2 inputs while that better measurement existed. The honest statement: this registration postdates JUNO’s *first* release (whose data are folded into Appendix B) and predates the *sub-percent JUNO dataset* (~2027–2028), which is the pre-declared deciding measurement; the forward-test character of the registration attaches to that dataset only; (3) until a third-party deposit (Zenodo) of this note and the ε -computation receipts exists, the registration has no independent timestamp and **no evidential standing beyond this repository’s own dating** — we say so plainly, and the deposit is the named next step.

Appendix A: change record (kept cumulative, against our interest)

From the 2026-06-10 internal version:

1. The strict-form tension was quoted as 2.5σ against $\sqrt{\varepsilon}$; this is inconsistent with the program’s own spectrum (it sets $m_1 = 0$ while simultaneously asserting $m_1 = \varepsilon m_3$). The self-consistent prediction is $\sqrt{\varepsilon/(1+\varepsilon)}$ and the tension is 1.4σ . The correction *reduces* the claimed tension; it is made because it is right, not because it helps.
2. The registered coefficient band $C \in [0.80, 1.25]$ is withdrawn (it contradicted the program’s own coefficient spread).
3. The inverted-ordering arithmetic of the earlier version is deleted (the formula is meaningless under IO); IO is now an explicit voiding scenario.
4. The look-elsewhere accounting is extended from the strict form to the band form and the trials count.

From the first publishable draft (after independent hostile review):

1. The registered claim was rebuilt: ”the exponent $x_\nu = 1/2$ ” with a free coefficient is *unfalsifiable* (the 68% coefficient spread covers 101% of the exponent axis), so the registration now names the undressed spectrum point, which is falsifiable and is the program’s actual assertion.
2. The expected JUNO outcome ($\approx 5.6\sigma$ exclusion as then computed at the rounded constant; 5.5σ at the exact constant per the Section 2 correction — historical record, kept as written; exclusion at current central values — death) was computed and stated; the previous draft omitted it.
3. The trials record was corrected: m_e/m_μ (exponent 1.546) *is* a band-grade hit of rung 1.5; the per-trial probabilities were unified to the self-consistent target (2.5% observed-grade; 25.9% band-grade; ≥ 2 -of-3 band accident 16.6%); constant-menu freedom is now counted.
4. The running escape hatch was closed with a computed SM bound ($\lesssim 5 \times 10^{-5}$, matching scale declared) and the $(2\sigma, 5\sigma]$ outcome band was pre-assigned (failed).
5. Σm_ν arithmetic corrected (0.0607 eV, 1.9 meV above the 0.0587 eV floor — the earlier ”1.5 meV” was wrong); ”fit correlation neglected, conservative” demoted to ”direction

unverified” — an overstated error bar *understates* tension, which is anti-conservative for an agreement claim.

The successor claim (cross-declaration)

Full disclosure that an earlier draft omitted: the program separately holds a *mechanism-derived dressed point* — $\sqrt{\varepsilon_{\text{eff}}/(1 + \varepsilon_{\text{eff}})} = 0.172747$ with $\varepsilon_{\text{eff}} = \varepsilon(1 - \varepsilon)$ — registered in the companion mechanism paper [IX] and its registration document, currently at -0.05σ against JUNO’s first data — and [IX]’s realization in fact carries a *two-member live menu* (the rung point 0.172747 and a cheaper texture at 0.167880, -3.0σ now), so the program holds three live points in total, all cross-declared. The successor’s chronology is disclosed there (the value postdates that data). The claims are mutually exclusive ($\sim 4\sigma$ apart at JUNO-final) and *at most* one can survive (data near 0.170 fails all three — this note’s point at $> 5\sigma$, [IX]’s members in their failed bands); the reader of this note must know that this note’s expected failure does *not* leave the program clean — it transfers the burden to [IX]’s claim, whose own death criteria are listed there. A registration pair structured this way is only honest if both documents cross-declare, which they now do.

Appendix B: status update against JUNO’s first data (2026-06-11)

JUNO’s first measurement (59.1 days; arXiv:2511.14593; *Nature* **654**, 343 (2026)): $\Delta m_{21}^2 = (7.50 \pm 0.12) \times 10^{-5} \text{ eV}^2$ (normal ordering) — already $1.6\times$ more precise than all previous data combined. With Δm_{31}^2 still from the global fit:

$$S_\nu = 0.17283 \pm 0.00167 \quad (\text{was } 0.1718 \pm 0.0026),$$

so the registered point (0.175472 at exact ε) now sits at $+1.6\sigma$ (was $+1.4\sigma$); equivalently, the registered point requires $\Delta m_{21}^2 = 7.73 \times 10^{-5}$, which is 1.9σ above JUNO’s measured value. Two things happened at once and we report both: the central value **moved toward the prediction** (from 0.1718 to 0.1728: 28% of the full gap to the registered point, 44% of the way to its 2σ survival boundary — both denominators stated) while the error bar shrank, leaving the significance nearly unchanged. Neither pre-committed threshold of Section 3 has been crossed; the deciding dataset remains the sub-percent JUNO measurement ($\sim 2027\text{--}2028$), at which point the $(2\sigma, 5\sigma]$ failure band and the $> 5\sigma$ kill apply as registered. The Section 2–3 numbers are preserved as the registration snapshot; this appendix is the dated update, and the companion program paper (corpus Paper 34) maps which texture coefficients remain compatible ($C_{\text{needed}} = 0.985 \pm 0.009$ against this data).

Computation record

```
eps = 0.031768636446582 exactly (= 3 kappa - 1; kappa =
    0.343922878815527, closed form in companion [C0])
registered point: sqrt(eps/(1+eps)) = 0.175472 (x_pred = 0.5045;
    the registration snapshot printed 0.17556 at the 3-digit
    rounding eps = 0.0318 - see Section 2's correction)
```

NO: $S = \sqrt{7.41e-5 / 2.511e-3} = 0.17179 \pm 0.00260$
 $x_{\text{measured}} = 0.5107$; $|x - x_{\text{pred}}| = 0.0062$
 $\text{deviation today} = (0.175472 - 0.17179)/0.00260 = 1.4 \text{ sigma}$
 JUNO expectation:
 $\sigma_S \sim 0.5 * \sqrt{0.006^2 + 0.005^2} * 0.1718 \sim 0.00067$
 at snapshot central: $0.00368/0.00067 \sim 5.5 \text{ sigma}$ (kill zone)
 at JUNO-first central: $0.00264/0.00067 \sim 4.0 \text{ sigma}$ (failed band)
 Sum of masses: $m_3 = \sqrt{2.511e-3/(1 - \epsilon^2)} = 0.050135 \text{ eV}$
 $\text{Sigma} = m_3 (1 + \sqrt{\epsilon} + \epsilon) = 0.0607 \text{ eV}$
 NO floor ($m_1 = 0$): $0.008609 + 0.050110 = 0.0587 \text{ eV}$
 running bound (SM, $\Lambda \leq 1e12 \text{ GeV}$):
 $y_{\tau}^2/(16 \pi^2) * \ln(\Lambda/m_Z) \sim 6.6e-7 * 23 \sim 1.5e-5$,
 with $O(1)$ flavor factors $\sim 5e-5$ relative on m_2/m_3
 (error propagation: half the quadrature sum of the relative mass-squared errors; fit correlation direction unverified)

References

- I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, T. Schwetz, A. Zhou, NuFIT 5.2 (2022), www.nu-fit.org — oscillation inputs of the registration snapshot (normal ordering, without SK-atm).
- JUNO collaboration, First measurement of reactor neutrino oscillations at JUNO, arXiv:2511.14593; *Nature* **654** (2026) 343 — the first JUNO data of Appendix B.
- JUNO collaboration, *Prog. Part. Nucl. Phys.* **123** (2022) 103927 — projected sub-percent precision; the resolving dataset.
- C0 *Exponential-consistency fixed points of parity-relation codes* (same author; same deposit) — the constant’s self-contained pedigree.
- IX *Scaled-seam suppression in Froggatt–Nielsen-type models* (same author; same deposit) — the successor claim set.
- P The program corpus (Papers 7–38, including Paper 34’s texture enumeration), archived as supplementary material with this deposit; [P] pointers are stated dependencies, checkable given the supplement.